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Implementing a multi-centric cohort study during the SARS-CoV-2 pandemic in a resource-constrained setting: challenges, adaptive solutions, and lessons from Brazil

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Abstract

Background Conducting multicenter, longitudinal epidemiological studies on emerging pathogens in real time during health emergencies presents significant challenges, especially in low- and middle-income countries, where substantial logistical and resource constraints hinder the rapid design and implementation of research protocols. This is particularly difficult when the disease spreads rapidly and its nature changes due to the onset of vaccination.

Methods During the COVID-19 pandemic, a cohort study (AVISA) was conducted in eleven research sites to assess seroprevalence. The objective of this study is to identify the challenges to the successful execution of the AVISA study through a retrospective analysis of quantitative and qualitative indicators.

Results The rapid spread of SARS-CoV-2 infections, particularly in large urban areas in Brazil, and the initiation of COVID-19 vaccination in January 2021 posed significant challenges for recruiting individuals who met the eligibility criteria. Participant enrollment was slower than initially planned, and fewer participants from the youngest and oldest cohorts were recruited. The implementation of the study protocol encountered significant logistical and operational challenges due to the evolving COVID-19 context, including the effects of vaccination, the loosening of social distancing measures, and difficulties conducting household visits and collecting samples in high-risk areas. To ensure participant retention and protocol adherence, the study employed flexible strategies, including rescheduling visits, enhancing communication, and mobilizing experienced staff, which were crucial for maintaining data collection and study continuity over time. Despite concurrent demands of the COVID-19 response, Instituto Butantan and its partners successfully mobilized resources, staff, and infrastructure to conduct the AVISA study across 11 sites,

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leveraging existing collaborations from prior clinical trials. The study conducted over 16,000 household visits, collected nearly 7,000 blood samples, and gathered detailed clinical and sociodemographic data from participants.

Conclusions Despite the complexities of conducting real-time research during a public health emergency, the AVISA study demonstrated that large-scale, multi-site longitudinal studies can be implemented in resource-constrained settings through adaptive strategies, institutional partnerships, and flexible field operations. The lessons learned offer essential insights for strengthening the design and execution of epidemiological studies in future health emergencies, particularly in low- and middle-income contexts.

Text box 1. Contributions to literature

- **Addresses Knowledge Gaps:** While large-scale surveillance is critical during epidemics, there is a lack of research on the operational challenges of implementing multi-center, real-time, dynamic studies.
- **Focus on Resource-Limited Settings:** This work underscores the urgent need for interventions to improve the speed and reliability of evidence generation within the specific constraints of low- and middle-income countries (LMICs).
- **Practical Application:** By detailing the implementation of a national longitudinal cohort in Brazil, this study provides a roadmap for assessing SARS-CoV-2 incidence and risk factors in complex, large-scale environments.

Background

The COVID-19 pandemic caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) has profoundly disrupted healthcare systems worldwide, straining medical infrastructure, overwhelming hospitals, and challenging public health responses [1]. Beyond its direct impact on morbidity and mortality, the pandemic has exacerbated pre-existing health disparities, disproportionately affecting vulnerable populations, including low-income communities, elderly individuals, and those who depend on public healthcare services [2, 3]. Socioeconomic inequalities, crowded living conditions, and occupational exposures have contributed to a greater burden of infection and adverse outcomes in these groups, underscoring the need for epidemiological studies that capture the real-time evolution of the pandemic across diverse demographic and socioeconomic contexts [4].

In response to the urgent need for robust epidemiological data to guide public health policies, the AVISA (Assessment of the Incidence of SARS-CoV-2 Infection and COVID-19 in Brazil [*Avaliação de incidência de infecção por SARS-CoV-2 e de COVID-19 no Brasil in Portuguese*]) Study was established as a multicentric longitudinal cohort study to assess the incidence and risk factors of SARS-CoV-2 infection and COVID-19 in Brazil [5]. Conducted in eleven municipalities across different regions (see Supplementary Materials), the AVISA Study generated an extensive dataset that enabled in-depth analysis of infection dynamics, risk factors, and the pandemic's broader health and social consequences. The study was led by the Instituto Butantan (IB), which is one of the largest manufacturers of vaccines and serum

in Brazil [6] and an institute that played a critical role in providing the first COVID-19 vaccine (Coronavac, Sinovac Biotech Ltd.) to the Brazilian Ministry of Health during the COVID-19 pandemic [7].

Conducting epidemiological studies on emerging pathogens in real time presents significant challenges [8, 9]. These include the need for rapid design and implementation, given logistical constraints in data collection and the pathogen's evolving nature. The objective of this study is to discuss the challenges and lessons learned from the AVISA Study in conducting a multicenter longitudinal cohort during the COVID-19 pandemic. We discuss the complications encountered in the study's implementation during the last months of the first wave of the COVID-19 pandemic (October to December 2020), when social distancing restrictions were in place, and then throughout the 14 months following the introduction of the COVID-19 vaccination (January 2021 to April 2022) [10, 11]. By analyzing the study's methodological, logistical, and operational aspects, we aim to provide insights into the feasibility of real-time epidemiological research during health emergencies. Additionally, we highlight the importance of adaptive strategies to overcome barriers to data collection, participant retention, and risk assessment in a rapidly evolving public health crisis.

Methods

This is a descriptive study based on qualitative and quantitative indicators, challenge descriptions, and reflections on the execution of the AVISA study. Following completion of the study, a retrospective analysis was conducted to identify the main implementation challenges during the COVID-19 pandemic. To validate the findings, the research team conducted debriefing meetings with the principal investigators and the fieldwork team at each study site, as well as with the investigators responsible for overseeing the study at IB.

Multi-site study design and sampling strategy for participant enrollment

The AVISA study aimed to assess the prevalence and incidence of SARS-CoV-2 infection and COVID-19 among populations served by the Family Health Strategy (Estratégia da Saúde da Família - ESF), the primary care program of the Unified Health System (Sistema Único da

Saúde). Under the coordination of the IB, eleven study sites were selected from four of the five Brazilian regions (Southeast, Northeast, North and Central-West). Each site enrolled participants and was prepared to conduct follow-up for 72 weeks in accordance with the original clinical protocol.

Based on the list of households enrolled in the Family Health Strategy, AVISA used a two-stage probabilistic cluster sampling approach to select households with micro areas selected in the first stage and households in the second stage. In the two smaller municipalities, where the number of micro-areas was less than 150, only households were randomly selected. All residents in the randomly selected households in the sample were invited to participate in the study.¹

Residents of each household across nine age groups, from the youngest to the oldest, who agreed to participate in the study, were followed with monthly monitoring visits. The sample size calculation for participant recruitment was based on the work of Hanley and Lippman-Hand et al. [12], considering nine age groups with baseline prevalence levels adjusted by 5%.

Different methods were used to determine the sample size for participants above and below the 50-year-old cut-off. In the youngest five age groups [(0–9), (10–19), (20–29), (30–39) and (40–49)], sample size estimates were calculated assuming type I error equal to 5% (two-tailed), an estimated risk of 10% for symptomatic SARS-CoV-2 infection and a 1% risk of hospitalization. For older adults (aged 50 years and older), the target recruitment sample was increased, given that the estimated risk of hospitalization increases with age, as described by Verity et al. [13]. Hospitalization risk was assumed to be 8.16% for the 50–59 cohort, 11.80% for the 60–69 cohort, 16.60% for the 70–79 cohort, and 19.40% for the 80–89 cohort. The final sample size was calculated to be 3,520 participants across 11 study sites, with approximately 320 participants expected to be enrolled at each site. This calculation included a 20% adjustment for potential dropout. Sampling weights were adjusted for post-stratification using population characteristics for each study municipality, accounting for the municipality's micro-area and the nine age strata.

¹ To conduct the random selection, a list of all existing micro-areas and their corresponding size measures (number of households) was obtained from each municipality. From this list, 30 micro-areas were randomly selected, with probability proportional to size, and within these, replicates of 90 households were selected. The replicates were used until the sample was complete for each age group. To compensate for changes in draw probability resulting from varying numbers of replicates, post-stratification weights were calculated after participant recruitment to adjust for age distribution imbalances between the study sample and the target population in each study municipality.

Data collection procedures and serological testing protocol

Households selected for participation and successfully recruited were visited at home by the research team. These visits were primarily conducted at the participants' homes but could also occur at the coordinating reference center. During the initial visit, household members' ages were recorded, and all individuals were invited to participate in the study. Participation in other clinical studies did not preclude individuals from volunteering for this study. However, participants who had received pharmacological or immunobiological prophylaxis for COVID-19 were ineligible for inclusion in the AVISA study.

Health teams collected individual demographic data, including medical history and household profile. Weight, height, temperature, and oxygen saturation were measured during the first visit. A questionnaire regarding behaviors related to potential exposure, participants' symptoms, and clinical care was also collected. Participants were considered enrolled in the study once blood samples for rapid and serological SARS-CoV-2 testing were collected during the first visit.

Rapid serological tests (COVID-19 IgM/IgG test, immunochromatographic assay) were performed every four weeks. In addition, venous blood samples were collected every 12 weeks after study enrollment or whenever the rapid test yielded a reactive IgM result for confirmatory testing using ECLIA (Elecsys® anti-SARS-CoV2 S e N). Participants who tested positive on serological assessments were to be tested at each subsequent visit to evaluate the duration of the antibody response. Venous blood samples were sent monthly by research centers to the Strategic Laboratory for Diagnosis of Instituto Butantan (LED-IB) in São Paulo, Brazil, for analysis. The sequential use of both rapid and laboratory-based serological tests aimed to increase the positive predictive value, given the expected low prevalence in the study population. The protocol defined exclusion criteria for individuals receiving pharmacological or immunobiological prophylaxis products for COVID-19 or SARS-CoV-2 infection.

The original study protocol was designed at the onset of the COVID-19 pandemic health emergency (see Supplementary Materials). At the time, there was limited knowledge about the novel pathogen's short- and medium-term dynamics, or about whether effective treatments or vaccines would be developed. To ensure adaptation and compliance across geographically dispersed teams, regular meetings and training sessions were held to discuss trade-offs in protocol changes and to ensure their effective implementation.

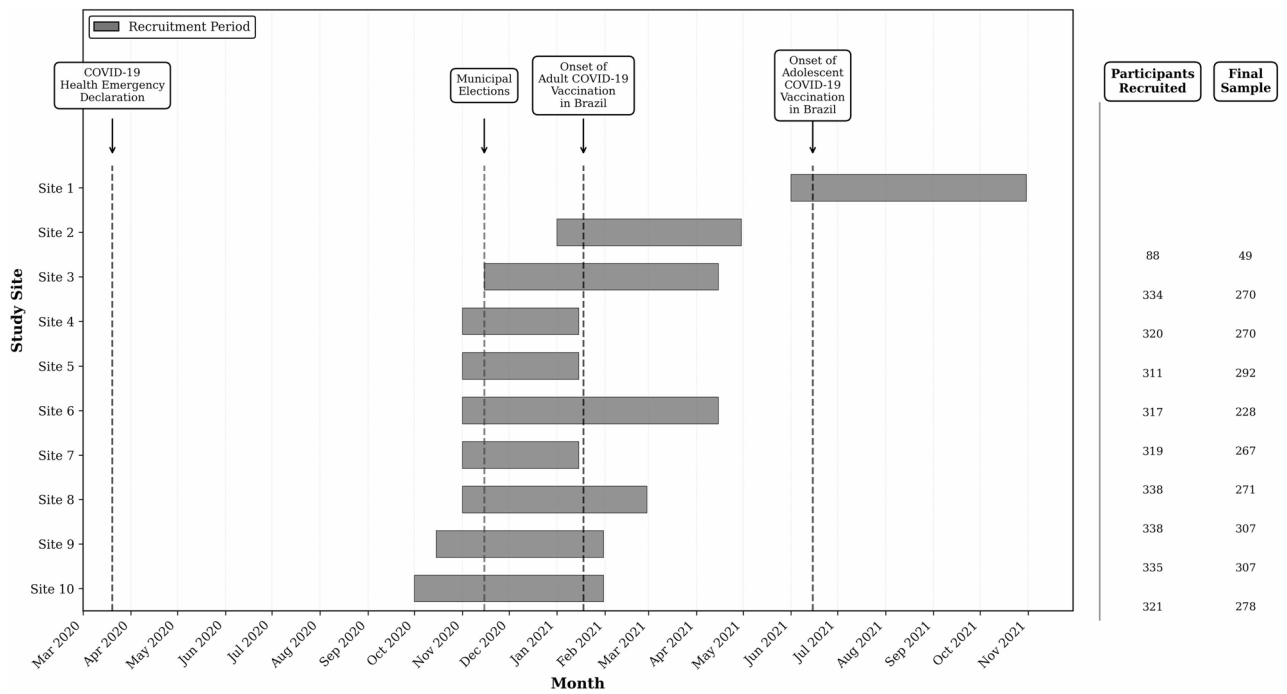


Fig. 1 Enrollment Periods, Key Events, and Recruitment Outcomes Across AVISA Study Centers

Results

Several challenges were encountered during the study procedures. We categorize these challenges into three domains: (a) participant enrollment and cohort composition; (b) implementation of the study protocol; and (c) mobilization resources to conduct the study.

Participant enrollment and cohort composition

Due to heterogeneous progress in securing site agreements and ethics approvals, the study started at different time points across sites, with the earliest centers initiating recruitment in October 2020. Participant recruitment concluded on October 5, 2021, with an average recruitment period of approximately 3.5 months per center. Most of the recruitment period occurred during the first wave and the early second wave of the COVID-19 pandemic, when the Alpha and Gamma variants were predominant [10]. A significant proportion of the study population tested positive for SARS-CoV-2 during the initial intake visit [14]. Figure 1 shows the key events concerning the COVID-19 pandemic in Brazil and at the AVISA study sites, as well as the duration of recruitment and the total number of participants recruited at each center.

COVID-19 vaccination began in Brazil on January 17, 2021 [15], while recruitment for the AVISA study was still ongoing. Vaccination initially targeted populations at high risk of developing severe COVID-19, including elderly people; and additional groups were systematically included as more vaccines became available. As the

Table 1 Participant exclusions in the AVISA study

Exclusion Stage	Number Excluded (n)
Excluded at screening	12
Excluded following verification of inclusion criteria	77
Excluded due to missing baseline test results	2
Total excluded	91

national vaccination campaign progressed, recruiting participants became progressively more difficult, particularly among older adults who were prioritized to receive doses due to their elevated risk of severe COVID-19 disease and death, thus reducing the pool of eligible unvaccinated participants.

A total of 3,077 individuals were initially recruited, but 91 were excluded (see Table 1). Twelve individuals were excluded at screening. Following verification, an additional 77 participants were excluded, including all 44 participants from a specific site due to quality issues and the remainder due to exclusion criteria. In the final sample, two additional participants were excluded due to missing baseline test results.

Table 2 reports the target population by age group and the effective sample recruited per site. The final study population consisted of 2,986 individuals (15.2% below the target sample size), drawn from approximately 700 households. Among age groups, the study had difficulty recruiting the youngest (children under 10 years) and the oldest (adults aged 79 and older). The remaining groups

Table 2 Recruitment targets and achievement by age group in the AVISA study

Age Group (years)	Target Sample (n)	Recruited Sample (n)	Recruitment Shortfall (n)	Recruitment Achievement (%)
0–9	392	257	135	65.6
10–19	452	409	43	90.5
20–29	494	476	18	96.4
30–39	512	457	55	89.3
40–49	410	392	18	95.6
50–59	481	407	74	84.6
60–69	332	302	30	91.0
70–79	235	173	62	73.6
80+	212	113	99	53.3
Total	3,520	2,986	534	84.8

Recruitment achievement percentages calculated as (Recruited Sample/Target Sample) × 100. The youngest (0–9 years) and oldest (70–79, 80+ years) age groups showed the lowest recruitment achievement rates

showed minor differences between the target population and the effectively recruited sample. Only two study sites did not reach the expected sample size.

Implementation of the study protocol

COVID-19 vaccination also posed challenges to completing the study protocol, particularly given the evolving emergency context. As vaccination coverage increased among participants and vaccine uptake was high among the target study population, several adjustments were necessary. First, the IgG/IgM serological rapid test was discontinued for participants who had already started COVID-19 vaccination. Second, the follow-up period was reduced from the originally planned 72 weeks to 60 weeks due to the rapid increase in vaccination. Third, the continued monitoring of antibodies of SARS-CoV-2 positive serology cases was discontinued. Additionally, information on vaccination status was included in the study protocol, and each individual's vaccination record was updated through the final study visit.

Considering the social distancing protocols in place [11], the logistical challenges in organizing staff and transportation for household visits were substantial. Both staff and the participants were trained to follow safety protocols to reduce transmission risks, and considerable efforts were undertaken to ensure adherence to these protocols throughout the 60-week follow-up period. Initially, stricter social distancing policies were enforced across most municipalities, and most home visits were successful in ensuring that the required data were collected. However, as social distancing mandates were relaxed and more individuals resumed activities outside their homes, the logistical challenges of conducting home visits increased, often requiring multiple attempts to complete data collection. In addition, given

the socio-economic vulnerability of this population, whose incomes were significantly below Brazil's national per capita income, many participants were unable to remain at home, which increased their mobility and created additional logistical obstacles for the study team. To overcome these difficulties, new strategies were adopted, such as scheduling regular visits at alternative times, including evenings and weekends, or at participants' workplaces during their lunch breaks.

Another common challenge was participants changing addresses during the follow-up period. In several cases, these changes were not communicated to the research team, or participants reported that the move was temporary, thereby allowing them to stay active in the research and continue to access regular testing. This situation created logistical difficulties in planning visits, particularly for individuals who relocated to other municipalities or states. As a mitigation strategy for the risk of follow-up loss, the study teams maintained active contact with participants by telephone, reinforcing the importance of updating their addresses and providing logistical support for the visits. All these measures, in addition to a continuous and active search effort and flexible appointment scheduling, were essential in maintaining participant retention throughout the study period.

The clinical monitoring of the study participants was also a significant challenge. It was planned to be conducted remotely and based on predefined risk assumptions. Even so, with monitoring personnel concurrently engaged in both the AVISA study and critical pandemic response activities, study monitoring was delayed, necessitating retrospective data cleaning rather than real-time cleaning and an adaptation of the initial monitoring plan. Local monitoring visits were conducted to ensure the quality of essential data was maintained and that the study adhered to research ethical norms.

Although other diagnostic tests could be used to investigate SARS-CoV-2 infection among participants, this study routinely used serological tests, including IgG/IgM immunochromatographic assays and ECLIA. As a result, it was not possible to differentiate, by serological methods, between antibodies generated by natural infection and those induced by vaccination. Furthermore, because serological rapid tests were unsuitable for assessing infection incidence among previously vaccinated individuals, their use in the study was discontinued in February 2022. Thus, it was necessary to censor the individual's contribution to the study analysis until their first infection or receipt of their first vaccine.

The logistics of venous blood sample transport presented ongoing challenges. Samples collected during household visits were transported to local study sites and then shipped to the central laboratory (LED-IB) for processing. Managing this multi-site sample transport chain

required continuous coordination, and transportation procedures had to be tailored to each site's specific logistical constraints. Although the protocol recommended sending samples monthly from each site to São Paulo, this schedule was not consistently adhered to due to limited flight availability during the COVID-19 pandemic.

Field-based insights suggest that collecting samples in households presented a significant challenge for study centers and their staff. Many residencies were in high-risk urban areas, raising concerns about the safety of field teams and materials. Additionally, many households lacked sufficient space to collect blood samples safely and appropriately.

However, the study's routine visits and testing enabled participants to obtain health services and access COVID-19 testing during the pandemic. The experience of receiving monthly tests created dissatisfaction when the research team discontinued rapid testing for individuals who had received their first dose. Many participants from socioeconomically vulnerable backgrounds expressed concerns about the discontinuation of these tests. To maintain participant retention, research teams consistently worked to ensure that communications encouraged continued participation in the study, even without the provision of a monthly rapid test.

To overcome these challenges, various operational strategies were implemented, such as pre-scheduling home visits at safe times; getting support from community health agents who accompanied the team during visits, improving access and enhancing safety; assigning professionals with greater agility and experience to perform blood collections; and, whenever possible, assigning male professionals for visiting high-risk urban areas. However, no additional security measures were provided for male professionals. These actions were crucial in ensuring the continuity of laboratory procedures, as outlined in the study protocol. In summary, a well-prepared and flexible team was essential to the successful implementation of the study in the field.

Mobilizing resources to conduct the study

The study protocol mandated monthly home visits for serological testing, with venous blood samples collected every 12 weeks and finger pricks performed at all visits. Responses to a structured questionnaire were also conducted during monthly visits. Household visits required logistical coordination and equipment, as well as sufficient staff to perform all study procedures on-site and in the field. Study personal contracts, equipment, tests, and materials required to perform all procedures were provided by the study sponsor (IB) with financial support from two non-profit organizations (UMANE and Instituto Todos pela Saúde).

Strategically, the study sites were quickly selected considering those sites where IB had already worked in previous vaccine clinical trials. Most of them were conducting another clinical trial in partnership with IB, which facilitated the administrative procedures required to engage investigators and their institutions, hire on-site staff in a timely manner, and provide essential local resources to initiate the study as quickly as possible. Nonetheless, delays were observed in certain steps of the process at several study sites.

For example, while the AVISA study planning and implementation was in execution, the IB was intensively working on its COVID-19 vaccine Phase 3 clinical trial and supporting the health system with COVID-19 surveillance and diagnosis within the São Paulo state territory.

Notwithstanding other pressing activities critical to mitigating the spread of the COVID-19 pandemic, the IB and its partner institutions across the ten sites successfully coordinated their efforts. The research teams from the 10 sites conducted 16,021 study visits and reported 15,965 rapid serology test results. A total of 6,895 venous blood samples were collected for central serology processing by the ten study centers (see Fig. 2).

Discussion

The AVISA Study provides valuable insights into the multifaceted impact of the COVID-19 pandemic on public health systems, particularly in middle-income countries. By examining a diverse cohort over an extended period, the study offers a comprehensive understanding of how different demographic groups and individuals with varying health conditions have been affected. While serosurvey studies conducted at specific time points have offered valuable insights into transmission dynamics in particular localities [16–22], the AVISA study is unique in its longitudinal design, following cohorts of households over an extended period across four regions of Brazil. This approach enabled researchers to address additional questions and gain a deeper understanding of evolving patterns within families and across households in diverse settings. This type of research is instrumental in shaping future public health strategies and policies, ensuring that they are tailored to address the specific needs and challenges identified. Furthermore, the findings underscore the importance of adaptive and resilient health systems that can effectively respond to pandemics and other health crises. As the world continues to grapple with the ongoing effects of COVID-19, the lessons learned from the AVISA Study will be crucial in guiding both immediate and long-term health interventions.

To gain a comprehensive understanding of the urgent need for robust epidemiological data to inform public health policies, this article elucidates the challenges and

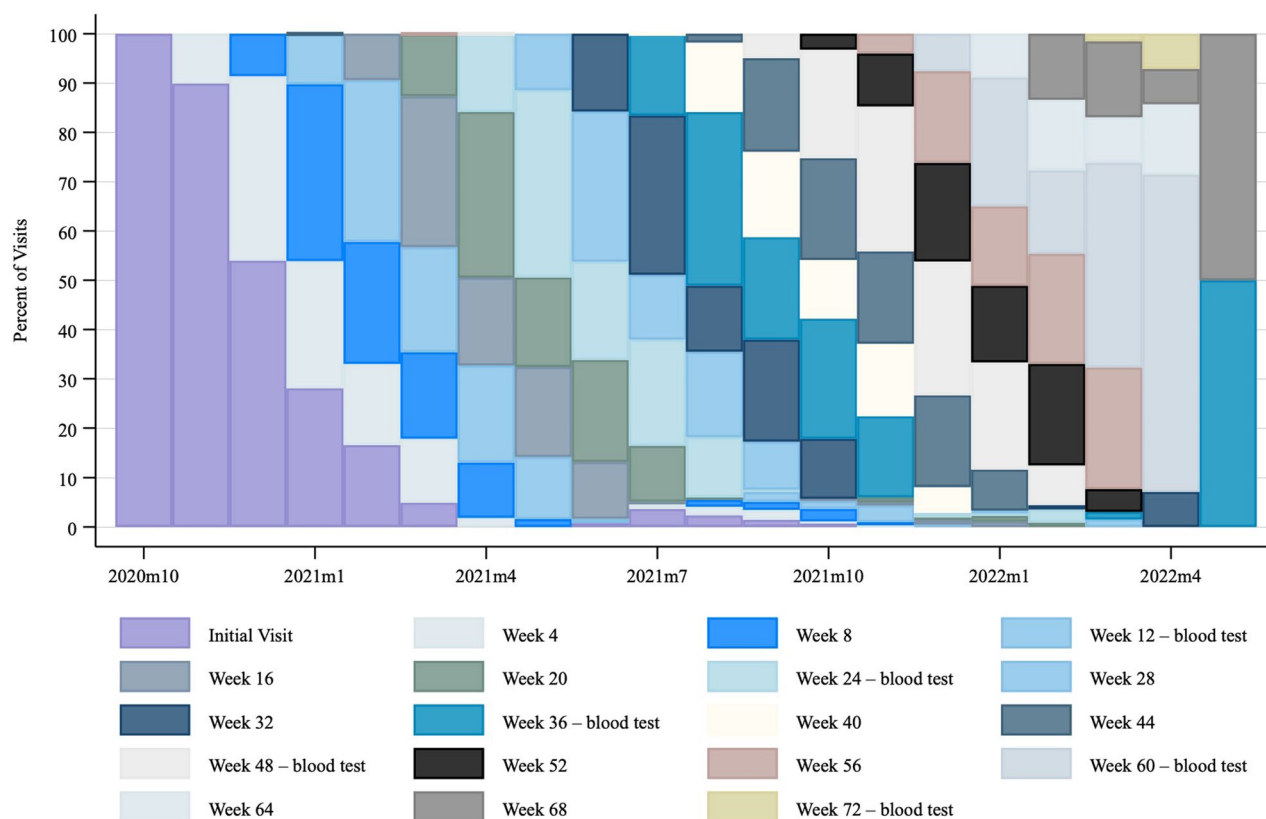


Fig. 2 Percent of Visit Type per Month across Avisa Sites

lessons learned from the AVISA Study, with a focus on Brazil's national COVID-19 response and the study's applicability to future pandemic preparedness. Based on discussions with AVISA study site investigators and coordinators, as well as the study management team, three primary themes emerged: participant recruitment and cohort composition, study protocol implementation, and resource mobilization for study execution. These themes are pivotal in shaping the public health response and enhancing future pandemic preparedness. The findings underscore several critical areas that warrant consideration in strategic planning to strengthen and reform public health.

Participant recruitment and cohort composition

Mitigating the risk of COVID-19 exposure posed significant challenges for authorities, particularly in low- and middle-income countries, including Brazil. At the onset of the COVID-19 pandemic, there was limited understanding of transmission dynamics within these contexts and their potential evolution throughout the pandemic [23]. Given the high demand for public health resources, it was crucial to comprehend the extent of household exposure to SARS-CoV-2 and the severity of the disease. This refined understanding is essential for developing

effective strategies to manage current and future public health crises.

Fewer elderly and younger individuals were recruited than initially planned. For the elderly, this may be attributed to their susceptibility, as they were recognized as particularly vulnerable to severe SARS-CoV-2 disease and had higher associated lethality [24]. Additionally, they were among the first groups to receive COVID-19 vaccines, which may have reduced their availability or willingness to participate. The lower recruitment of children was unexpected and more difficult to understand. While adults may have been more willing to participate in the study, they may have hesitated to enroll their children because of the considerable difficulty of obtaining blood samples from pediatric participants [25].

Recruitment challenges for the AVISA cohort study were compounded by the difficulty, using serological methods, of distinguishing between anti-SARS-CoV-2 antibodies generated by natural infection and those induced by vaccination when the study was implemented [26]. This limitation limited the analysis to data from participants through the time of their first infection or receipt of their first vaccine dose.

The challenges in achieving the study population composition for all age cohorts are consistent with findings from COVID-19 health surveys, indicating that certain

population groups are more easily recruited and more likely to remain engaged in follow-up studies [27–29]. This condition suggests that different recruitment approaches are necessary for specific age groups, particularly the youngest and oldest cohorts. For older and more vulnerable populations, efforts may need more time to recruit and to understand specific concerns. One of the benefits relayed by participants and health professionals who participated in the study was that home visits were important for building trust and facilitating access to health services during the health emergency. Accordingly, study protocols might be designed or modified to better address participants' concerns and needs [30].

Study protocol

During the COVID-19 pandemic, research methodologies had to be adapted. Studies have documented that research teams were obligated to swiftly modify their protocols to accommodate remote data collection [31] and virtual participant engagement [32], thereby ensuring the continuity of studies while adhering to safety measures. The AVISA study was designed considering the challenges posed by the COVID-19 pandemic. However, as conditions evolved, modifications to the study protocol became necessary (see Supplementary Materials). In this regard, the AVISA study underscores the importance of designing multicenter studies, as protocols must be sufficiently flexible to adapt to emerging challenges.

As study protocols evolved, digital data-collection tools were updated, necessitating ongoing training for field teams. The AVISA study highlights the value of integrating study protocols with digital platforms, as this integration can be instrumental in supporting protocol adaptation to changing conditions in an emergency setting. At the same time, as a multicenter study, AVISA highlights that online modules and webinars, continuous education and skill development need to be prioritized to ensure adaptation and compliance across geographically dispersed teams. The AVISA experience in using these tools not only kept all members current with protocol changes but also promoted a culture of learning and flexibility, both critical in the dynamic context of a pandemic response.

Resource mobilization

Our findings are consistent with other studies conducted in low and middle-income countries, highlighting the challenges caused by insufficient resources and inadequate logistics within the healthcare sector [33]. The systemic issues faced by healthcare systems in these resource-limited settings posed an essential challenge throughout the AVISA study. Considering limited resources, resource allocation decisions were pivotal in

strengthening local team capabilities to successfully complete the study.

In terms of logistics and supplies, the AVISA study highlighted that establishing partnerships with local suppliers and logistics companies can significantly streamline the acquisition and distribution of necessary materials [26, 30]. By leveraging local networks, research teams ensured timely access to essential supplies, reduced transportation costs, and mitigated risks associated with supply chain disruptions. Additionally, implementing robust inventory management systems contributed to tracking and managing resources efficiently, minimizing wastage and ensuring optimal utilization.

Community engagement strategies were enriched by collaborating with local leaders and organizations to co-create awareness campaigns [34]. By involving community stakeholders in the design and execution of engagement activities, research teams built trust and fostered a sense of ownership among participants. This collaborative approach enhanced the effectiveness of outreach efforts and strengthened the relationship between researchers and the community, leading to higher levels of participation and cooperation.

Agile and assertive communication, exemplified by the use of project management tools that facilitated real-time collaboration and information sharing among research coordinators, streamlined communication, tracked progress, and ensured that all team members were aligned with the study's objectives and timelines. Regular virtual meetings and updates further enhanced transparency and accountability, enabling swift decision-making and problem-solving.

The study's coordination among research centers was enhanced by organizing regular workshops and conferences that brought together experts from various fields to share insights and best practices. These events served as platforms for knowledge exchange, fostering innovation and collaboration across research centers. By creating a network of interconnected research hubs, teams leveraged collective expertise and resources to address common challenges and drive impactful solutions. Interdisciplinary collaboration was vital to addressing these challenges and integrating diverse expertise backgrounds to develop innovative solutions.

Maintaining a stable central technical team was crucial for ensuring consistency and continuity in research efforts [35]. This was achieved by implementing succession planning and mentorship programs that prepared junior team members for leadership roles. By nurturing talent and providing opportunities for professional growth, research centers built resilient and capable technical teams that adapted to changing circumstances and sustained the research initiative over the long term.

Conclusions

In summary, the AVISA Study provides valuable insights into the complex challenges associated with conducting epidemiological research in pandemic contexts. It highlights the necessity of flexible strategies and robust data collection methods to effectively inform public health responses. By expanding on these insights and incorporating practical examples, research teams can enhance their preparedness and resilience, thereby ensuring a comprehensive and effective response to future health crises. Overall, the AVISA Study provides a foundation for building resilient public health systems, encouraging innovation and collaboration to navigate pandemic complexities and safeguard communities globally.

Abbreviations

ITpS	Instituto Todos pela Saúde
SARS-CoV-2	Severe acute respiratory syndrome coronavirus 2
AVISA Study	Assessment of the incidence of SARS-CoV-2 infection and COVID-19 in Brazil (AVISA Study) or in Portuguese, <i>Avaliação de Incidência de Infecção por SARS-CoV-2 e de COVID-19 no Brasil</i>
ESF	Family Health Strategy (FHS) or in Portuguese <i>Estratégia da Saúde da Família</i>

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13690-026-01870-x>.

Supplementary Material 1.

Supplementary Material 2.

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Authors' contributions

L.B., M.A.R., R.Q.G., V.S.S., and E.N.C.B. conceptualized and conducted the formal analysis and wrote the manuscript text. L.B. prepared all tables and figures. M.C.B. reviewed and edited the draft of the manuscript. Data curation, analysis and validation of the study's findings was provided by M.A.R., R.Q.G., V.S.S., E.N.C.B., M.C.B., G.R., A.J.d.F., M.M.T., M.L.N., C.J.F.F., I.C.B.C., D.B.P., A.M.d.S., D.H.R.S., E.G.P., M.T.R.P.C., M.M.E., and M.C.G.P.A. All authors reviewed the manuscript.

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Data availability

Data is provided within the manuscript or supplementary information files.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Committee of Hospital Leforte de São Paulo (CAAE: 31228620.1.1001.548) and by the local ethics committees at each participating study site. The study was conducted in accordance with the principles of the Declaration of Helsinki. Before inclusion, all participants or their legal representatives signed a written consent form.

Consent for publication

The authors consent to publication.

Competing interests

The authors declare no competing interests.

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